A robot that acts to know

Day 1 — The challenge, probability, and free energy

Course built on the doctoral thesis of Noé Zapata

Monday 30 November

Montevideo, 30 November – 4 December 2026

The challenge

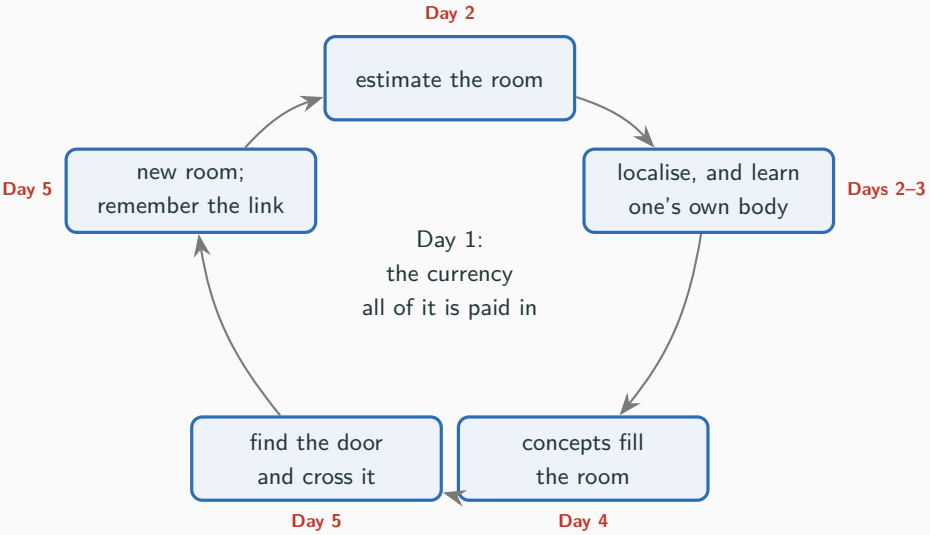
A robot is put down in an apartment it has never seen

[video: the full cycle, Webots / real apartment]

room estimated → localised → objects appear → door crossed → second room

Watch for one thing: nothing in this video was programmed as a sequence of steps.

The cycle we will build this week

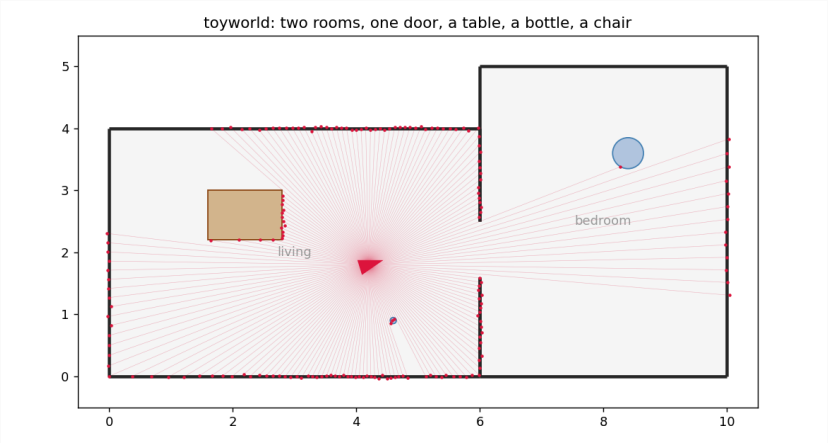


Every stage is a belief

stage	what is unknown (latent)	what supplies the evidence	day
estimate the room	walls, corners, the polygon	LiDAR returns	2
localise	the robot's pose in that room	LiDAR against the believed room	2
learn the body	wheel scale, gyro bias, sensor mounts	the localiser's corrections	3
concepts	which objects exist, where, what shape	camera masks, LiDAR	4
door	where it is, whether it is open	the room's boundary, free space	5
new room + memory	a second polygon; how rooms connect	all of the above, again	5

One generative model over all of it, and one quantity that every stage minimises: perception lowers it by changing beliefs, action by changing the data.

The toy version you will build, one stage a day



Two rooms, one door, three objects, a LiDAR, and a robot whose wheels lie a little. Pure `numpy`; it runs on your laptop.

How this course is built

Each session runs through three steps:

1. the **theory**, derived rather than asserted;
2. the **code**, one stage of the cycle, in the lab;
3. the **measurements** that showed when the equations were wrong.

The third step is not optional. Most of what the thesis learned, it learned there.

The thread through all five days

Averaging removes independent error.
It does not remove shared error.

You will meet it as:

- the common-mode ceiling;
- a prior re-centred on its own estimate;
- a fused precision better than its best member;
- a metric that scores what the robot never saw.

The grammar of uncertainty

What the robot is uncertain about

deterministic	$y = f(x)$: one outcome
random	$y = f(x) + \varepsilon$: many outcomes
discrete	a door is open *or* closed
continuous	a wall is 4.013 . . . m away

Everything the robot does not know is a **random variable**, and what it believes about it is a **probability distribution**: a table for discrete variables, a density for continuous ones.

$$s = \begin{bmatrix} x \\ y \\ \theta \\ w \\ d \end{bmatrix} \quad \begin{array}{l} \text{robot pose} \\ \text{room width, depth} \end{array}$$

$$o = \begin{bmatrix} p_1 \\ \vdots \\ p_N \end{bmatrix} \text{ LiDAR returns} \qquad a = \begin{bmatrix} v_{\text{adv}} \\ v_{\text{side}} \\ \omega \end{bmatrix} \text{ command}$$

The whole problem is one object:

$$p(s, o, a)$$

The joint distribution: everything there is to know

A two-variable world. D: is the door open? O: does the LiDAR see a gap where the door is?

	$O = \text{gap}$	$O = \text{wall}$
$D = \text{open}$	0.27	0.03
$D = \text{closed}$	0.14	0.56

the four numbers add up to 1

- The **joint** $p(D, O)$ gives the probability of every combination at once.
- Every question you can ask about D and O is a calculation on this table, with only three operations: summing, slicing, and multiplying back.
- Everything this week, including free energy, is these three operations, done on tables too large to write down.

Marginalisation: sum out what you do not care about

	gap	wall	$p(D)$
open	0.27	0.03	0.30
closed	0.14	0.56	0.70
$p(O)$	0.41	0.59	1

$$p(o) = \sum_d p(d, o) \qquad p(o) = \int p(s, o)\, \mathrm{d}s$$

- Sum each row or column and write the result *in the margin*: that is where the name comes from.
- A marginal answers a question about one variable while staying honest about every value the other one might take.
- The integral over s is the expensive one. Remember it.

Conditioning: keep the slice you observed, renormalise

	gap	wall
open	0.27	0.03
closed	0.14	0.56
$p(O)$	0.41	0.59

$$p(\text{open} \mid \text{gap}) = \frac{0.27}{0.41} = 0.66$$

$$p(s \mid o) = \frac{p(s, o)}{p(o)}$$

- Observing $O = \text{gap}$ throws away the column that did not happen. What remains no longer sums to 1, so divide by its total, the marginal $p(o)$.
- Before looking, the door was open with probability 0.30; after seeing a gap, 0.66. Conditioning is how evidence enters a belief.
- Seeing a wall instead: $0.03/0.59 = 0.05$.

Factorisation: the product rule

Turn the conditioning formula around:

$$p(s, o) \;=\; \underbrace{p(o \mid s)\, p(s)}_{\text{first } s\text{, then } o \text{ given } s} \;=\; \underbrace{p(s \mid o)\, p(o)}_{\text{first } o\text{, then } s \text{ given } o}$$

Check it on the table:

- $p(\text{gap} \mid \text{open})\, p(\text{open}) = 0.9 \times 0.30 = 0.27$
- $p(\text{open} \mid \text{gap})\, p(\text{gap}) = 0.66 \times 0.41 = 0.27$

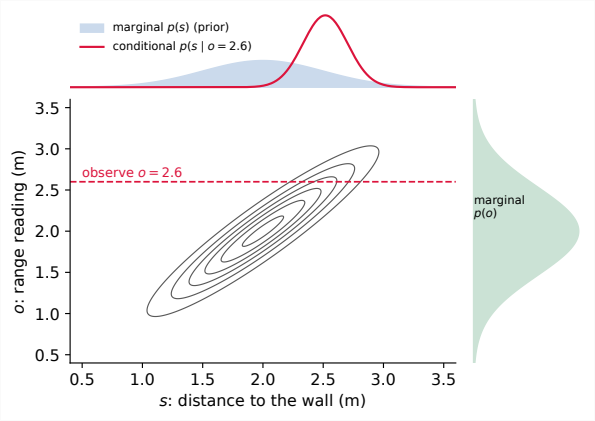
Both products rebuild the same cell of the joint.

Set the two factorisations equal and divide by $p(o)$:

$$p(s \mid o) = \frac{p(o \mid s)\, p(s)}{p(o)}$$

Bayes' rule is not an extra axiom. It is the product rule written twice.

The same operations on continuous variables



s: distance to a wall, prior $\mathcal{N}(2, 0.5^2)$.
o: its range reading, $o \mid s \sim \mathcal{N}(s, 0.2^2)$.

- The **joint** is the tilted ellipse: readings track distances.
- A **marginal** is the shadow the joint casts on one axis.
- A **conditional** is one horizontal cut through it, rescaled to area 1.

Reading $o = 2.6$ moves the belief from 2.00 ± 0.50 to 2.52 ± 0.19 m.

source: figures/make_day1.py, block 6

For Gaussians, both operations are linear algebra

Split a Gaussian over (s, o) into blocks, written two ways:

$$\text{covariance } \Sigma = \begin{bmatrix} \Sigma_{ss} & \Sigma_{so} \\ \Sigma_{os} & \Sigma_{oo} \end{bmatrix} \qquad \text{precision } \Lambda = \Sigma^{-1} = \begin{bmatrix} \Lambda_{ss} & \Lambda_{so} \\ \Lambda_{os} & \Lambda_{oo} \end{bmatrix}$$

	in covariance form	in precision form
marginalise out o	keep the block: Σ_{ss}	Schur complement: $\Lambda_{ss} - \Lambda_{so}\Lambda_{oo}^{-1}\Lambda_{os}$
condition on o	Schur complement: $\Sigma_{ss} - \Sigma_{so}\Sigma_{oo}^{-1}\Sigma_{os}$	keep the block: Λ_{ss}

- Each form makes one operation free and the other costly. The slide before: $0.25 - 0.25^2/0.29 = 0.035 = 0.19^2$.
- The robot keeps its beliefs in precision form. When it forgets an old pose (Days 2–5), it is marginalising, and it pays a Schur complement.

The chain rule: N variables, N factors

Apply the product rule again and again, peeling off one variable each time:

$$
\begin{aligned}
p(A, B) &= p(A \mid B)\, p(B) \\
p(A, B, C) &= p(A \mid B, C)\, p(B \mid C)\, p(C) \\
p(A, B, C, D) &= p(A \mid B, C, D)\, p(B \mid C, D)\, p(C \mid D)\, p(D)
\end{aligned}
$$

In general, for any ordering π of the variables:

$$
p(X_1, \ldots, X_N) = \underbrace{p(X_{\pi(1)})\, p(X_{\pi(2)} \mid X_{\pi(1)}) \cdots p(X_{\pi(N)} \mid X_{\pi(1)}, \ldots, X_{\pi(N-1)})}_{N \text{ factors}}
$$

- No assumption was made. This is exact for every distribution.
- But the last factor depends on everything: nothing has been gained yet.

One joint, $N!$ factorisations

From now on the names we use all week: S the **state** (is the door open?), O the **observation** (does the LiDAR see a gap?), A the **action** (does the robot turn its LiDAR towards the door, or away?).

$$
\begin{aligned}
p(S, O, A) &= p(S)\, p(O \mid S)\, p(A \mid S, O) \\
&= p(S)\, p(A \mid S)\, p(O \mid S, A) \\
&= p(O)\, p(S \mid O)\, p(A \mid S, O) \\
&= p(O)\, p(A \mid O)\, p(S \mid O, A) \\
&= p(A)\, p(S \mid A)\, p(O \mid S, A) \\
&= p(A)\, p(O \mid A)\, p(S \mid O, A)
\end{aligned}
$$

- 3 variables give $3! = 6$ orderings, each with 3 factors.
- N variables give $N!$. A robot tracked over 100 steps has hundreds of variables.
- All of them are exactly equal. No amount of data can tell them apart.

So which one do we write down?

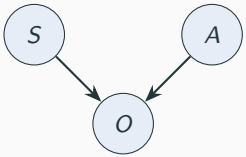
Choose the order in which the world produces things

Causal order: causes before effects.

$$p(S)\,p(A \mid S)\,p(O \mid S, A)$$

The robot cannot see whether the door is open, so where it points its LiDAR does not depend on it; what it observes depends on both:

$$p(A \mid S) = p(A)$$



Anti-causal order: effects first.

$$p(O)\,p(A \mid O)\,p(S \mid O, A)$$

Once O is known, S and A are linked through it (a gap means little if the LiDAR looked away), so nothing simplifies. Every factor keeps all its arguments.

Numbers needed to specify it (binary variables):

any order, nothing dropped	$2^3 - 1 = 7$
causal, using $A \perp S$	$1 + 1 + 4 = 6$
N variables, any order	$2^N - 1$
N variables, $\leq k$ causes each	$\leq N\,2^k$

Why the causal order wins

1. **Sparse.** Each variable depends on a few direct causes, not on everything before it. The arrows you leave out are the independences you know.
2. **Each factor is a piece of physics.** $p(O \mid S, A)$ is a sensor model and $p(s_t \mid s_{t-1}, a_t)$ a motion model; you can write them down. Nobody can write down $p(O)$, the distribution of every possible LiDAR scan.
3. **Modular.** Replace the LiDAR with a camera and only $p(O \mid S, A)$ changes. In the anti-causal order, all three factors change.
4. **It runs forwards.** Sampling the factors in order *generates* data: the model can predict what it will see if it acts. Action needs exactly that.

Three things you can build from one joint

Perception only

$$p(s \mid o)$$

From an image, guess the state: a classifier, a detector.

It answers one question, in one direction.

Control only

$$p(a \mid o)$$

From an observation, choose the action: a PID loop, a hand-written controller, a policy learnt by reinforcement.

It is the factor $p(A \mid O)$ of the anti-causal order, used on its own.

The generative model

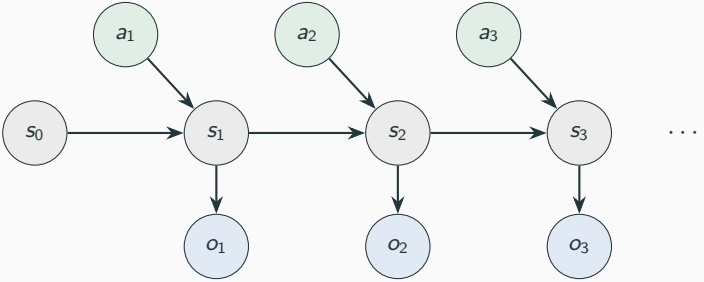
$$p(s)\, p(a)\, p(o \mid s, a)$$

Explain what is seen, predict what each action would show, and choose the action by inference.

This is what we build.

A controller or a learnt policy can act very well, in the world it was tuned or trained in. What it cannot do is say what it expects to see, notice that it is surprised, tell what it does not know, or survive a sensor swap without being re-tuned: none of that is in $p(a \mid o)$. Every one of those is a question the joint can answer.

The robot's joint, in causal order



$$p(s_{0:T}, o_{1:T}, a_{1:T}) = \underbrace{p(s_0)}_{\text{prior}} \prod_{t=1}^{T} \underbrace{p(a_t)}_{\text{policy}} \underbrace{p(s_t \mid s_{t-1}, a_t)}_{\text{motion model}} \underbrace{p(o_t \mid s_t)}_{\text{sensor model}}$$

$(3T+1)!$ orderings are available, and we write one: each factor touches at most three variables and is a piece of physics. This is the **generative model**; on Day 2 it becomes the factor graph the robot actually solves.

Bayes' rule is the three operations in a row

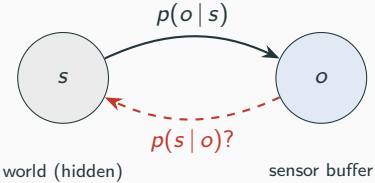
$$\underbrace{p(s \mid o)}_{\text{posterior}} = \frac{\overbrace{p(o \mid s)}^{\text{likelihood}}\ \overbrace{p(s)}^{\text{prior}}}{\underbrace{p(o)}_{\text{evidence}}}$$

operation	where	what it asks
factorise, causally	numerator	how would a robot at s see this room? what did it expect?
condition on o	left side	where is the robot, now that this scan has arrived?
marginalise s	denominator	how likely was this scan, averaged over every s?

The first two are cheap. The third is where exact inference dies,
and the rest of today is about how not to compute it.

Inference behind a sensory boundary

The agent never sees the state



The **generative model**:

$$p(s, o) = p(o \mid s)\, p(s)$$

- $p(o \mid s)$ is the agent's theory of its own sensor: which LiDAR returns a robot at this pose, in this room, would produce.
- $p(s)$ is what it brought with it: plausible room shapes, typical table sizes.

Inference runs the arrow backwards:

$$p(s \mid o) = \frac{p(o \mid s)\, p(s)}{p(o)}$$

source: active_inference.tex l.181–212

Where exact inference dies

$$p(o) = \int p(o \mid s)\, p(s)\, \mathrm{d}s$$

- The probability of *this scan* under *every* room and furniture configuration the robot considers possible, each weighted by how likely it is.
- No closed form and no feasible quadrature. This is not a quirk of one sensor: it is the generic situation.
- Every practical method from here on is a way of not computing it.

A word about "state"

s is a latent variable of the *agent's model*, not the physical truth, which is never accessible. That distinction is what gives the question "is this belief wrong?" its meaning.

source: active_inference.tex l.200–220

Variational free energy

Do not compute the posterior; approximate it

Choose a tractable family, and a member $q(s)$ of it (the **belief**). Move it towards the true posterior, measuring closeness by

$$D_{\mathrm{KL}}[q(s) \,\|\, p(s \mid o)] = \int q(s) \ln \frac{q(s)}{p(s \mid o)} \,\mathrm{d}s \;\geq 0.$$

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Problem: it contains $p(s \mid o)$, the very thing we cannot have.

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Problem: it contains $p(s \mid o)$, the very thing we cannot have.

Substitute Bayes' rule and separate out the part that does not depend on q:

$$D_{\mathrm{KL}}[q \,\|\, p(s \mid o)] = \underbrace{\int q(s) \ln \frac{q(s)}{p(o, s)} \,\mathrm{d}s}_{F[q,o]} \;+\; \ln p(o)$$

F needs only the belief and the joint $p(o \mid s)p(s)$ the agent already wrote down.

source: active_inference.tex l.222–250

The identity the framework rests on

$$F[q, o] = D_{\mathrm{KL}}[q(s) \,\|\, p(s \mid o)] - \ln p(o)$$

Reading 1: same minimiser

$\ln p(o)$ does not depend on q. Minimising the computable F minimises the uncomputable divergence.

Reading 2: a bound on surprise

$\mathrm{KL} \geq 0$, so $F \geq -\ln p(o)$. Lowering F makes the belief better *and* the data less surprising under the agent's own model.

If you know machine learning: $-F$ is the **ELBO**. Maximising the ELBO and minimising free energy are the same operation.

source: active_inference.tex l.252–268

Complexity minus accuracy

$$F[q, o] = \underbrace{D_{\mathrm{KL}}[q(s) \,\|\, p(s)]}_{\text{complexity}} - \underbrace{\mathbb{E}_q[\ln p(o \mid s)]}_{\text{accuracy}}$$

- **Accuracy** rewards beliefs under which the data are likely. On its own, it would adopt any explanation that fits the current scan.
- **Complexity** charges for the distance travelled from the prior: Occam's razor, written as a divergence.
- Free energy picks the belief that buys the most accuracy for the least departure from what was already believed.

One trade, stated once, in nats, for every quantity the agent believes.

source: active_inference.tex l.288–308

Worked example: the Kalman update is free-energy minimisation

Take the prior $s \sim \mathcal{N}(m, p)$, an observation $y \sim \mathcal{N}(s, r)$, and the belief $q = \mathcal{N}(\mu, \Sigma)$. Every term is a Gaussian expectation:

$$F(\mu, \Sigma) = \frac{(\mu - m)^2 + \Sigma}{2p} + \frac{(y - \mu)^2 + \Sigma}{2r} - \tfrac{1}{2} \ln \Sigma + \text{const}$$

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$\partial F / \partial \mu = 0$:

$$\mu = \frac{m/p + y/r}{1/p + 1/r}$$

the precision-weighted mean.

$\partial F / \partial \Sigma = 0$:

$$\frac{1}{\Sigma} = \frac{1}{p} + \frac{1}{r}$$

precisions add.

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The two halves come from different terms. The *mean* comes from the prediction errors; the *variance* comes from the curvature of F balanced against the entropy $-\frac{1}{2} \ln \Sigma$. Regularised least squares is the same trade with a flat prior.

A choice, marked as a choice

- Closeness was measured with KL. Nothing forced that: it is the $\alpha \to 1$ member of the Rényi family, and the bound survives the substitution.
- The order α controls the *character* of the error: does the belief err towards too broad, or towards too confident?
- On this robot, that is not a statistical nicety. A covariance is spent: on deciding where to look next, and on how far a measurement may move a belief.

The thesis keeps $\alpha \to 1$ throughout. Making α itself a latent variable is open future work.

source: active_inference.tex l.270–286; future_work.tex l.22

The Laplace approximation

Gaussian beliefs: the form everything later takes

Take $q(s) = \mathcal{N}(\mu, \Sigma)$ and expand around the mode:

$$F \approx -\ln p(o, \mu) + \tfrac{1}{2}\,\mathrm{tr}(\mathcal{H}\Sigma) - \tfrac{1}{2}\ln|\Sigma| - \tfrac{n}{2}\ln(2\pi e), \qquad \mathcal{H} = -\nabla_s^2 \ln p(o, s)\big|_{\mu}$$

- The **mean** moves to the mode of the joint: a nonlinear least-squares problem (Gauss–Newton).
- The **covariance** is pulled two ways: the curvature term shrinks it, the entropy term widens it. Setting $\partial F / \partial \Sigma = 0$:

$$\boxed{\Sigma = \mathcal{H}^{-1}}$$

The uncertainty comes as a by-product of fitting, at no extra modelling cost.

source: active_inference.tex l.310–333

Gauss–Newton, in one line

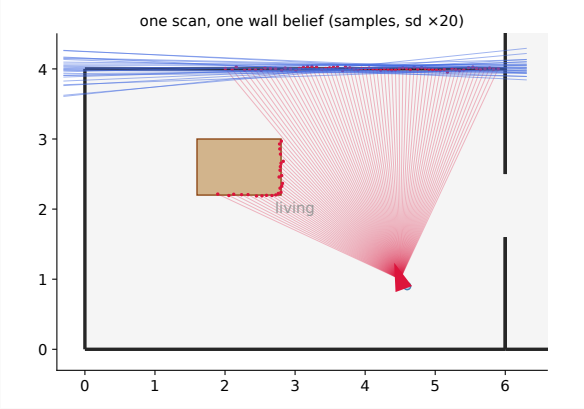
For whitened residuals $r(s)$ with noise covariance R, and a Gaussian prior $\mathcal{N}(m, P)$:

$$-\ln p(o, s) = \tfrac{1}{2} r^\top R^{-1} r + \tfrac{1}{2}(s-m)^\top P^{-1}(s-m) + \text{const},$$

$$\mathcal{H} \approx J^\top R^{-1} J + P^{-1}, \qquad \Delta s = -\mathcal{H}^{-1}\big(J^\top R^{-1} r + P^{-1}(s-m)\big).$$

- Iterate to convergence. The final $\mathcal{H}^{-1}$ *is* the belief's covariance.
- In the lab, `laplace(residual, x0, sigma, prior=...)` does exactly this. It also returns the Laplace free energy $F \approx -\ln p(o, \mu) + \frac{1}{2}\ln|\mathcal{H}| - \frac{n}{2}\ln 2\pi$, which you will use to compare models.

Example: one wall as a belief



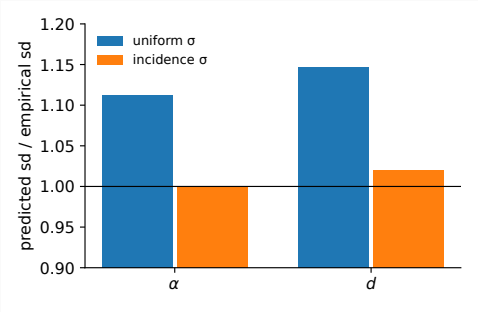
A line with normal $n(\alpha)$ at distance d:

$$r_i = \cos\alpha\, x_i + \sin\alpha\, y_i - d$$

$$\frac{\partial r_i}{\partial(\alpha, d)} = \big[\, -\sin\alpha\, x_i + \cos\alpha\, y_i,\ -1 \,\big]$$

- The first column is a **lever arm**. Points far apart along the wall pin the angle.
- α and d come out correlated ($\rho \approx -0.97$), because d is measured from an origin the points are far away from.

Is the covariance honest?



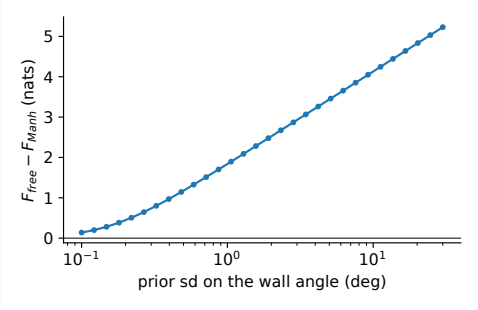
$\Sigma = \mathcal{H}^{-1}$ is a *claim* about how far the fit would move on a re-scan. Test it with 400 scans from one pose:

- uniform σ: about 10% too pessimistic, and it does not average away;
- the noise is along the *beam*, the residual along the wall's *normal*: $\sigma_i = \sigma\,|u_i^\top n|$;
- with that likelihood: within a few per cent.

A Laplace covariance is exactly as honest as the likelihood you wrote.

source: lab 1, exercises 3–4; the real localiser's incidence weight: room_concept.tex l.232

Free energy compares models: the price of a parameter



Two models of the same wall:

- **Manhattan**: the angle is fixed at 90°; one parameter, d.
- **free**: $\alpha \sim \mathcal{N}(90^\circ, s^2)$; two parameters.

Both fit equally well. The free model pays $\approx \ln(s/\sigma_\alpha)$ nats of *complexity*: the cost of squeezing the prior down to the posterior.

Tilt the points by 1.5° and the verdict flips by 70 nats.

Nobody wrote "if the angle error exceeds k degrees". That is the first instance of **no thresholds**.

What measurement said

Free energy is not computation

Measured: a better model does not save arithmetic

Injecting a body-model error from 0 to 10%:

- the solver iterations per correction stay flat to within a twentieth;
- the size of the correction grows by about half as much again.

- The agent pays in *how far its beliefs must move*. The arithmetic doing the moving is nearly constant.
- "A better model saves the agent work" has to say *which* work. The two readings come apart, and an endpoint chosen on the wrong reading measures nothing.

source: active_inference.tex l.835–848

A factor inherits the errors of what it is estimated through

- The body is learnt from the localiser's corrections, which are fitted against the room. So a wrong map is charged to the wheels.
- A robot in a room whose map is wrong will conclude that its wheels are wrong.
- And it gets *quieter*, not louder: once the body has absorbed the map's distortion, the two agree. The state is internally consistent and externally wrong.

Measured: an omitted parameter

Admitting a camera-mount translation the data cannot support moves the reported pitch by **nine times** its stated interval. An interval speaks only about the parameters the model contains.

Carry an external reference. Never trust only what the agent computes about itself.

source: active_inference.tex l.899–950

Lab

Lab 1: probability by hand, then a wall as a belief

`labs/day1.ipynb`, about 50 minutes. Today the robot knows its pose and we cheat on which points are the wall's.

1. **Warm-up**: the door table by hand (marginalise, condition, both factorisations); read an anti-causal order off a joint; condition a Gaussian in both forms.
2. Write the wall's residual and Jacobian; fit it with `laplace`.
3. **Honesty test**: 500 scans; predicted vs empirical σ. Find the 10% and fix the likelihood.
4. **Model choice**: Manhattan vs free, by ΔF. Then tilt the wall.

Where to look next waits for Day 3, once action has a functional of its own.

Tomorrow

The robot loses both of today's cheats:

- it no longer knows its pose;
- it no longer knows which points belong to which wall.

One wall becomes a room, the pose becomes one more belief, and the two must be estimated *together*.

And we measure the common-mode ceiling: how little a thousand points can teach when they share one error.

Reading for Day 2

Thesis chapters:

- *Estimating the Layout of an Unknown Room*, §§1–5;
- *The Room Concept Agent*, §§"What is estimated" and "The observation model".